

United Arab Emirates University (UAEU)
College of Engineering
Electrical and Communication Engineering Department

Record of Academic Achievements Submitted for Promotion to the Rank of Associate Professor

By

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1 INTRODUCTION AND BACKGROUND

Dr. Mohammad Al Bataineh, born in 1979 in Irbid, Jordan, began his academic journey at the Hijjawi Faculty for Engineering Technology, Yarmouk University, where he graduated with high honors, earning a Bachelor of Science in Telecommunications Engineering in 2003. Recognized for his academic excellence with consistent placement on the Dean's List, he went on to serve as a Teaching Assistant at Yarmouk University. This role earned him a scholarship to pursue advanced studies in the United States. At the Illinois Institute of Technology (IIT), Dr. Al Bataineh completed his Master's degree in Electrical Engineering in 2006, graduating with a perfect 4.0 GPA, and later earned his Ph.D. in Electrical Engineering in 2010, again maintaining a perfect 4.0 GPA. His doctoral research focused on applying communications, coding theory, and information theory to biological systems, particularly gene expression. During his doctoral studies, he gained valuable practical experience as a Research Assistant at Argonne National Laboratories and as a Research and Development Engineer at MicroSun Technologies in Illinois. In September 2010, Dr. Al Bataineh returned to Yarmouk University as an Assistant Professor in the Telecommunications Engineering Department, where he served until 2018, including a tenure as Department Head following his promotion to Associate Professor. In August 2020, Dr. Al Bataineh joined the United Arab Emirates University (UAEU) as an Assistant Professor, where he continues to contribute significantly to teaching, research, and service.

Research and Publications: Dr. Al Bataineh's research portfolio spans both theoretical and applied aspects of telecommunications engineering and bioinformatics. At Yarmouk University, he led innovative research in telecommunications, contributing significantly to the field through over 30 peer-reviewed publications, including journal articles and conference papers. His work has been published in high-impact journals such as IEEE Access and Sensors, with some papers achieving impact factors as high as 4.42. Since joining UAEU in 2020 and particularly in 2024-2025, Dr. Al Bataineh has significantly enhanced his research profile, publishing in high-impact journals including the *Journal of King Saud University - Engineering Sciences* (CiteScore: 12.1), *IEEE Open Journal of the Communications Society* (Impact Factor: 6.1, CiteScore: 10.5), *Cluster Computing* (Impact Factor: 4.1), *IEEE Access* (Impact Factor: 3.6, CiteScore: 9.0), and *IET Smart Cities* (Impact Factor: 3.1). His total publication count has grown to over 40 peer-reviewed articles, with improved citation metrics and h-index performance. Upon joining UAEU in 2020, Dr. Al Bataineh continued to push the boundaries of his field, securing substantial research funding, including a 367,000 AED start-up grant for his project on gene and regulatory sequence identification using communications theory. His role as Principal Investigator in multiple funded projects, such as the SDG and SURE Plus grants, underscores his ability to lead complex, interdisciplinary research initiatives that bridge engineering with cutting-edge biomedical applications.

Teaching Excellence: Throughout his academic career, Dr. Al Bataineh has demonstrated a strong commitment to teaching and student mentorship. He has taught a diverse array of courses at both Yarmouk University and UAEU, ranging from foundational undergraduate courses like Signals and Systems, Analog Communications, and Digital Image Processing to advanced graduate courses such as Digital Signal Processing, Information Theory and Coding, and Artificial Neural Networks. His dedication to pedagogical innovation is evidenced by his participation in the Faculty Teaching Academy Program (FTAP) and his role as Digital Liaison for the College of Engineering, leading blended course transformation initiatives aligned with modern instructional design principles. His dedication to education is further evidenced by his supervision of numerous GP projects and his active involvement in SDG and SURE Plus projects, where he has guided students through hands-on research experiences that prepare them for successful careers in engineering.

Service and Leadership: In addition to his research and teaching accomplishments, Dr. Al Bataineh has made significant contributions to university service. At UAEU, he has expanded his service portfolio to include roles as Chair of the Department Travel Committee, Technical Program Committee Member for IEEE AICT 2025, and Organizing Committee Administrator for IEEE BioCAS 2025. At UAEU, he is an active member of several departments and university committees, including the department-level assessment committee. He also serves as an academic advisor to 39 undergraduate students in the Electrical and Communications Engineering department, providing mentorship and guidance to help them achieve their academic and professional goals. His previous experience as Department Chair at Yarmouk University, where he led the department for two years, further illustrates his leadership capabilities and his commitment to advancing the field of telecommunications engineering through both academic and administrative roles.

2 TEACHING AND INSTRUCTIONAL ACTIVITIES

2.1 T1.1. Student evaluation for courses taught each semester

I have consistently received high evaluations for my teaching in most of the courses I have taught at UAEU and prior to that. In the average of my students' evaluation at UAEU, I achieved a score of 4.52 out of 5, which is more than the Department's average score of 4.43 (as of Spring 2024). Please see all student evaluation reports, student testimonials, and sample student gratitude emails in Appendix T1.1.1. Some of my student evaluations at Yarmouk University are also included in Appendix T1.1.2.

Detailed presentation of the course evaluations is presented below:

Table 1: Course Evaluation Scores

Term	Course	FA	DA	CA	UA
Fall 2020	Probability and Random Processes (ECOM 320-OL1)	4.62	4.33	4.32	4.40
	Digital Image Processing (ELEC 551-51)	4.47	4.33	4.32	4.40
	Digital Image Processing (ELEC 551-01)	4.12	4.33	4.32	4.40
Spring 2021	Signals and Systems (ELEC 360-01)	3.69	4.39	4.37	4.44
	Signals and Systems (ELEC 360-02-ZMC)	4.85	4.39	4.37	4.44
	Signals and Systems (ELEC 360-51)	4.35	4.39	4.37	4.44
Fall 2021	Probability and Random Processes (ECOM 320-51)	5.00	4.49	4.43	4.50
	Probability and Random Processes (ECOM 320-01)	4.58	4.49	4.43	4.50
	Digital Image Processing (ELEC 551-01)	4.66	4.49	4.43	4.50
	Digital Image Processing (ELEC 551-01)	4.73	4.49	4.43	4.50
Spring 2022	Signals and Systems (ELEC 360-01)	4.75	4.28	4.42	4.52
	Signals and Systems (ELEC 360-02)	4.55	4.28	4.42	4.52
	Information Theory and Coding (ECOM 561-01)	3.67	4.28	4.42	4.52
Fall 2022	Signals and Systems (ELEC 360-01)	4.53	4.63	4.54	4.54
	Signals and Systems (ELEC 360-02)	4.90	4.63	4.54	4.54
	Analytical Methods for Electrical Engineering (ELEC 380-01)	4.39	4.63	4.54	4.54
Spring 2023	Information Theory and Coding (ECOM 561-01)	4.69	4.40	4.56	4.56
	Information Theory and Coding (ECOM 561-51)	4.83	4.40	4.56	4.56
Fall 2023	Signals and Systems (ELEC 360-51)	4.46	4.45	4.55	4.54
	Signals and Systems (ELEC 360-01)	4.12	4.45	4.55	4.54
	Analytical Methods for EE (ELEC 380-01)	4.62	4.45	4.55	4.54
Spring 2024	Information Theory and Coding (ECOM 561-01)	NRP	4.43	4.51	4.55
	Artificial Neural Networks (ELEC 644)	4.79	4.43	4.51	4.55
Fall 2024	Signals and Systems (ELEC 360-01)	4.63	4.45	4.52	4.56
	Signals and Systems (ELEC 360-02)	4.42	4.45	4.52	4.56
	Analytical Methods for EE (ELEC 380-01)	4.96	4.45	4.52	4.56
Spring 2025	Signals and Systems (ELEC 360-01)	4.31	4.55	4.51	4.57
	Analytical Methods for EE (ELEC 380-01)	4.62	4.55	4.51	4.57
	Analytical Methods for EE (ELEC 380-02)	4.36	4.45	4.43	4.50
	Capstone Eng. Design Project (ELEC 590-51)	5.00	4.55	4.51	4.57
Average		4.54	4.44	4.47	4.51

The detailed analysis of the course evaluations, based on the provided data, is available in Appendix T1.1.3 for further review and insights. This analysis includes a comprehensive breakdown of the student feedback across various terms and courses. This feedback reflects my commitment to adapting my teaching to student needs and creating an engaging learning environment. (See Samples of student testimonials and student appreciation emails are provided in Appendix T1.1.1.)

2.2 T1.2. Peer evaluation at the department/program (and/or College) Level, consisting of a series of classroom visits.

To be provided by the PET committee.

2.3 T1.3. Teaching loads assigned to the faculty member and the diversity of courses he/she can deliver

My teaching load since joining UAEU is summarized in table 2 below. Please see Appendix T1.3.1 for supporting documents on the teaching load.

Table 2: Teaching load at UAEU

Term	2020-2021	2021-2022	2022-2023	2023-2024	2024-2025
Fall	11	13.5	9	13	12.5
Spring	11	10.5	10	7.5	11
Summer	0	3	0	0	3
Total	22	27	19	20.5	26.5

The list of courses I have taught at UAEU is as follows:

1. ECOM 320 – Probability and Random Processes
2. ELEC 551 – Digital Image Processing
3. ELEC 360 – Signals and Systems
4. ECOM 561 – Information Theory and Coding
5. ELEC 380 – Analytical Methods for Electrical Engineering

6. ELEC 644 – Artificial Neural Networks (Graduate level course)
7. ELEC 585 – Graduation Project - I
8. ELEC 590 – Graduation Project - II

Prior to joining UAEU, I have taught several other courses at Yarmouk University:

1. CME 312 – Signals and Systems
2. CME 314 – Probability and Random Processes
3. CME 452 – Digital Communications
4. CME 456 – Communications Systems
5. CME 612 – Digital Signal Processing (Graduate level Course)
6. CME 616 – Information Theory and Coding (Graduate level Course)

Please see Appendix T1.3.2 for syllabi of courses taught at Yarmouk University.

My diverse teaching load, from foundational to advanced courses, has enriched my instructional skills and allowed me to adapt my teaching methods to different learning levels, fostering a dynamic educational environment.

2.4 T1.4. Methodology and documentation of the teaching process, including the development of course plans and the achievement of course outcomes

- During the first class session, I thoroughly explain the course syllabi, which are available on Blackboard, and make a concerted effort to engage and motivate students about the course topics. I ensure that students understand the overarching theme and how it aligns with their broader study plans. I clearly communicate the expected learning outcomes, which are documented in the course syllabus and included in the slides for the initial session of every course I teach. This strategy fosters an environment where students are well-prepared and motivated from the outset (refer to Appendix T1.4.2 for supporting documents, including sample teaching slides and materials).
- From the beginning of each course, I consistently inform students about the course's general theme, its relevance to their academic goals, and the expected learning outcomes. These outcomes are clearly documented in the course syllabus and lecture slides, ensuring that students have a clear understanding of what is expected of them. Each course is assessed through various evaluation tools—such as quizzes, assignments, projects, midterm, and final exams—to comprehensively measure learning and ensure robust assessment (Appendix T1.4.2 and T1.4.3 contain sample materials and assessed assignments).
- I heavily depend on solving in-class worksheet problems to help students understand complex problems, design effective solutions, and develop problem-solving skills. Additionally, my courses include in-class quizzes, homework assignments, and exams to reinforce learning and assess understanding (Please refer to Appendix T1.4.3 for samples).
- I utilize technology to enhance learning by recording all lectures through Collaborate Ultra or MS Teams, ensuring students have continuous access to the content. Each section has a dedicated MS Teams class where all course materials—including syllabi, PowerPoint slides, assignments, and worksheets—are systematically uploaded to both MS Teams and Blackboard, providing a cohesive and comprehensive resource platform for students (see Appendix T1.4.4 for supporting materials).
- To foster engagement and deepen understanding, I actively involve students in discussions during lectures by posing thought-provoking questions. I also connect theoretical content to concepts familiar to students, which enhances their engagement and makes the material more relatable and interesting.
- All courses are meticulously documented in line with the university's procedures, using LOAMS—UAEU's internal system—to manage assessments and measure the achievement of learning outcomes. This systematic approach ensures transparency and accuracy in evaluating both teaching effectiveness and student learning (see Appendix T1.4.1 for sample assessment reports).
- I maintain comprehensive e-Portfolios for all UAEU courses, containing assessment instruments, reports, student work samples, and complete course materials. This systematic documentation supports continuous improvement and aligns with UAEU's teaching portfolio evaluation requirements (see Appendix T1.4.5 for e-Portfolio archives).
- My teaching methodology has been enhanced through the Faculty Teaching Academy Program (FTAP), where I implemented Constructivist Learning Theory (CLT) principles using evidence-based approaches including flipped classroom models, SMART learning objectives, and inclusive assessment strategies. This training strengthened my ability to design and document teaching processes that achieve course outcomes while promoting equity and meeting diverse learning needs (see Appendix T1.4.6 for FTAP certification and training materials).

2.5 T1.5. Development of course content

- **ELEC 644 (Artificial Neural Networks)**: Updated all course lectures and created comprehensive PowerPoint presentations with enhanced visual content. Introduced new materials covering deep learning architectures and practical implementation techniques, integrated MATLAB and Python coding examples. Content tailored for students from diverse majors and backgrounds. (See Appendix T1.5.1 for supporting materials.)

- **Undergraduate Courses at UAEU**: Created and refined course content for multiple undergraduate courses, including:
  - **Signals and Systems (ELEC 360)**
  - **Probability and Random Processes (ECOM 320)**
  - **Information Theory and Coding (ECOM 561)**
  - **Digital Image Processing (ELEC 551)**
  - **Analytical Methods for Electrical Engineering (ELEC 380)**

  These materials are designed to enhance understanding and engagement among students. (See Appendix T1.5.2 for supporting materials.)
- **Courses at Yarmouk University (YU)**: Developed course content for all undergraduate and graduate courses taught, including:
  - **Undergraduate Courses:** Signals and Systems (CME 312), Probability and Random Processes (CME 314), Digital Communications (CME 452), and Communication Systems (CME 456).
  - **Graduate Courses:** Digital Signal Processing (CME 612), Information Theory and Coding (CME 616).

  Adapted content to meet the learning needs at different educational levels. (See Appendix T1.5.3 for supporting materials.)
- **Tempus Project at YU**: Designed course descriptors and materials for various Master's-level courses under two tracks—Wireless Communications and Wireless Technology. This initiative was part of the Tempus project titled "An Industry-Oriented MSc. Program in Telecommunications Engineering—Towards an EU Approach," aiming to align with European educational standards. (See Appendix T1.5.4 for supporting materials.)

2.6 T1.6. Use and development of modern teaching methods and techniques

- **Interactive Lectures with iPad**: I always use my iPad to hand-write lectures, making them interactive and visually engaging. These notes are converted to PDFs and shared via Blackboard and MS Teams. The iPad also enables the use of apps like GeoGebra to visualize complex functions in 2D and 3D. (See Appendix T1.6.1.)
- **Digital Classrooms and Recorded Lectures**: I set up an MS Teams class for each course and record all lectures, allowing students to review material at their convenience. (See Appendix T1.6.2.)
- **Course Material Accessibility**: I post all course slides on Blackboard at the start of the term, encouraging students to review materials before class, which fosters more in-depth discussions and critical thinking during lectures. (See Appendix T1.6.3.)
- **Hands-On Learning with MATLAB**: I provide hands-on sessions using MATLAB and share sample codes in MATLAB and Python, promoting practical application and deeper understanding. (See Appendix T1.6.4.)
- **Supplemental Learning via YouTube**: My recorded lectures from Yarmouk University are available on YouTube, offering additional support for students to master course content. (See Appendix T1.6.5 or Signals and Systems, and Digital Signal Processing)
- **24/7 Communication via WhatsApp**: I create a WhatsApp group for each course, enabling continuous communication and support, allowing students to reach out with questions and engage in discussions outside of class hours. (See Appendix T1.6.6.)
- **Constructivist Learning Theory Implementation**: Through my participation in the Faculty Teaching Academy Program (FTAP), I have implemented Constructivist Learning Theory (CLT) principles in course development, utilizing evidence-based pedagogical approaches including flipped classroom models, SMART learning objectives, and inclusive assessment strategies (see Appendix T1.6.7 for CLT implementation examples and FTAP certification).
- **Digital Liaison for Educational Technology Integration**: Appointed as Digital Liaison for the College of Engineering (Spring 2025), supporting faculty in adopting emerging educational technologies and best practices in digital learning through coordination with CETL's course transformation initiatives (see Appendix T1.6.8 for appointment and coordination activities).
- **NotebookLM for Multi-Format Learning**: I use NotebookLM to convert lectures into podcasts, mental maps, and video overviews, which I then share with students. This approach supports different learning preferences and provides flexible access to course material. (See Appendix T1.6.9 for examples and proofs.)

  These methods enhance engagement, promote critical thinking, and support diverse learning styles.

2.7 T1.7. Contribution to development of academic programs, curricula and courses

- **New Program Proposal and Development at UAEU**: Served as a committee member for the proposal of the "Biomedical Engineering and Informatics" program. Developed the study plan for the "Biomedical Instruments and Systems" course after benchmarking with international programs. (See Appendix T1.7.1.)
- **Course Content Development at UAEU**: Developed and implemented new content for several courses, including Artificial Neural Networks (ELEC 644), Signals and Systems (ELEC 360), Probability and Random Processes (ECOM 320), Digital Image Processing (ELEC 551), Analytical Methods for Electrical Engineering (ELEC 380), and

Information Theory and Coding (ECOM 561) to enhance curriculum alignment with student needs and modern practices. (See Appendix T1.7.2.)

- **Chaired ECOM Program Analysis Committee at UAEU**: Chaired a committee to analyze the ECOM program using PRAMS data, assessing course offerings, program viability, and financial health. Prepared a report outlining current standings and recommending remedial actions for improvement. (See Appendix T1.7.3.)
- **Course Revision Committee for Communications Engineering Program at UAEU**: Served as a committee member involved in revising the Communications Engineering Program, contributing to updates and improvements in course content and structure. (See Appendix T1.7.4.)
- **ECE Department Assessment Committee at UAEU**: Served as a member of the assessment committee from AY 2021 to 2025, actively participating in the evaluation and enhancement of departmental courses and academic standards. (See Appendix T1.7.5.)
- **Master's Program Curriculum Development at YU**: Led the design of Master's-level courses under the Tempus project, aligning with industry standards and EU educational frameworks to enhance program quality. (See Appendix T1.7.6.)
- **Programs Update Committee Participation**: Participated in multiple Programs Update Committees during AY 2024-2025, contributing to curricular review and program updates across the College of Engineering to ensure alignment with industry needs and accreditation standards (see Appendix T1.7.7 for committee reports and recommendations).

These contributions reflect my dedication to enhancing academic programs and curricula, ensuring they remain relevant and aligned with educational and industry standards.

2.8 T1.8. Preparation of appropriate examinations and the development of effective student evaluation tools, to support course objectives and to achieve course goals, and grade distribution curve

- I design examinations and evaluation tools (e.g., assignments, term projects, quizzes, class participation, midterm exams, and final exams) that align with course objectives, combining theoretical questions with practical problems to assess both knowledge and application. The exam questions vary in difficulty and are crafted to accurately reflect each student's understanding of the course material and its objectives. These tools include a variety of assessment methods such as quizzes, assignments, projects, and both formative and summative assessments, all supported by clear rubrics to ensure fair grading. I regularly adjust these tools based on student feedback to enhance their effectiveness in supporting course goals and improving learning outcomes. (See Appendix T1.8.1 for samples of exams and assessment tools.)
- Grade distributions across my courses demonstrate fair evaluation standards with appropriate performance spread, aligning with learning outcomes and maintaining academic rigor (see Appendix T1.8.2 for grade distribution analysis).
- Assessed students' technical writing proficiency through the evaluation of term project reports and graduation project reports.
- Monitored and evaluated the weekly progress of students enrolled in Graduation Projects I and II by systematically measuring their attainment of pre-defined goals and expected outcomes.
- **Quality Matters (QM) Training**: I am scheduled to complete the Quality Matters "Applying the QM Rubric" certification training on September 23, 2025, sponsored by CETL, which will further enhance my ability to design effective assessment tools aligned with industry-standard quality assurance principles (see Appendix T1.8.3 for QM training registration and curriculum details).

2.9 T1.9. Interactions with students and encouraging them to develop their skills and enhance their self-learning capabilities

- **Active Engagement**: I foster a supportive learning environment through regular interactions with students, encouraging open communication and active participation in class discussions. (See Appendix T1.9.1 for related documents.)
- **Skill Development**: I promote critical thinking and problem-solving skills by incorporating hands-on projects and real-world exercises in the curriculum. (See Appendix T1.9.2 for sample projects and exercises.)
- **Self-Learning Encouragement**: I provide resources such as sample codes, recorded lectures, and supplementary materials to help students explore topics independently and enhance their self-learning capabilities. (See Appendix T1.9.3 for shared materials and resources.)
- **24/7 Support**: I create WhatsApp groups for each course to provide continuous support and facilitate communication, allowing students to ask questions and collaborate outside of class hours. (See Appendix T1.9.4 for communication records.)
- **Research and Exploration**: I motivate students to engage in research projects and explore additional resources, fostering a culture of lifelong learning and intellectual curiosity. (See Appendix T1.9.5 for research projects and student feedback.)

- **Academic Advising**: I serve as an academic advisor for 39 students, offering continuous guidance tailored to their unique academic statuses and helping them navigate their educational journeys. (See Appendix T1.9.6 for advising logs and student testimonials.)
- **Office Hours**: I maintain regular office hours, both in-person and virtual, to provide accessible academic and personal support. (See Appendix T1.9.8 for office hours schedules and records.)
- **Judging Student Competitions**: I served as a judge at the IEEE UAE Student Day and Ibn Trais Award, where I evaluated student projects and provided constructive feedback on their designs. (See Appendix T1.9.9 for judging event details.)
- **SET Analysis**: I analyze SET feedback on participation (Q6) and critical thinking (Q7), with consistently high scores confirming effective student engagement. (See Appendix T1.9.10 for SET reports.)
- **Innovation Competition Organization**: I organized and hosted the INNOSPARK Award 2025, an innovation-focused student competition held on February 4, 2025. This event promoted student innovation, presentation skills, and teamwork while providing mentorship and networking opportunities. I coordinated logistics, facilitated sessions, and moderated key discussions to enhance students' creative and entrepreneurial capabilities (see Appendix T1.9.7 for event documentation and participant feedback).
- **Research-Integrated Student Mentorship**: Through supervision of SURE Plus Grant projects, I provide undergraduate students with hands-on research experience in cutting-edge technologies, fostering their research skills and preparing them for advanced studies. This mentorship extends beyond traditional classroom interaction to include research methodology training, project management, and professional development guidance. (See Appendix T1.10.4 and Appendix S1.7.10 for supervision records.)

2.10 T1.10. Supervision of hands-on training, graduation projects, laboratory activities and seminars

- **Supervision of Graduation Projects**: I have supervised several graduation projects at UAEU, focusing on innovative solutions such as **Wi-Fi Enabled, Remotely-Controlled Surveillance Car**, **Real-time Text-to-Speech Assistive Glasses for Visually Impaired Individuals**, and **Smart Assistive Eyewear for Enhanced Visual Support and Real-Time Face Recognition**. Currently (Fall 2024), I am supervising one GP I project and one GP II project. (See Appendix T1.10.1 for a snapshot from the Graduate Project Management system and a sample GPI and GPII reports.)
- **Supervision of Graduation Projects at Yarmouk University**: While at Yarmouk University, I supervised over 10 graduation projects, providing guidance on diverse engineering topics. (See Appendix T1.10.2 for samples of supervised projects.)
- **Hands-on Training and Industrial Training Supervision**: Since joining UAEU in 2020, I have consistently supervised students during their industrial training, helping them apply their theoretical knowledge in real-world industry settings and gain valuable professional experience. (See Appendix T1.10.3 for industrial training records.)
- **SURE Plus Grant Supervision**: Currently supervising SURE Plus Grant project "Offline Assistive System with Stereo Vision and Object Detection" (40,000 AED, May 2025–May 2026), providing undergraduate students hands-on research experience in computer vision and assistive technologies (see Appendix T1.10.4).

2.11 T1.11 Dissemination of teaching practices, methodologies etc. at the Departmental/Program, College or University levels

- **Sharing Course Content and Best Practices**: I share my course content with newly hired faculty members in the Electrical and Communication Engineering Department and provide them with guidance on best practices to enhance their teaching effectiveness. (See Appendix T1.11.1 for shared materials and communication records.)
- **YouTube Channel for Teaching Resources**: I created a YouTube channel featuring my in-class lectures for several courses taught at Yarmouk University. This resource is open to both students and colleagues to learn from and adopt effective teaching strategies. I have received significant positive feedback from both students at UAEU and Yarmouk University, as well as faculty members, on the valuable impact of my YouTube videos in enhancing their understanding and teaching practices. (See Appendix T1.11.2 for YouTube channel details.)
- **Sharing Innovative Teaching Methods**: I regularly share my teaching practices and methodologies, such as using iPad-based interactive lectures and integrating software like MATLAB and Python for hands-on learning with my colleagues at the ECE Department at UAEU. (See Appendix T1.11.3 sample communication.)
- **International Conference Submission on Educational Technology**: Submitted a research paper on technology-enhanced active learning methodologies to ICMET 2025, disseminating innovative teaching practices to the international engineering education community (see Appendix T1.11.4).
- **University Educational Innovation Talk**: Proposed presentation "Exponential vs. Linear Thinking: A Call to Rethink Education in the Age of Disruption" to disseminate innovative teaching methodologies, scheduled after September 15, 2025 (see Appendix T1.11.5).
- **International Educational Outreach via YouTube Channel**: Created and maintain a YouTube channel with 411+

educational videos and 10.8K subscribers, providing international dissemination of engineering education methodologies and innovative teaching practices. (See Appendix S1.12.1 for channel link and analytics and Appendix T1.11.2 for YouTube channel details.)

These activities demonstrate my commitment to enhancing teaching quality and fostering a collaborative educational environment within the department, college, and university.

2.12 T1.12. Honors and Awards for Teaching

- **Student Testimonials**: Although I have not received formal honors or awards, I have consistently received positive feedback and testimonials from students at both UAEU and Yarmouk University. These testimonials highlight my engaging teaching style, dedication to student success, and the impact of my methods on their learning experiences. (See Appendix T1.12.1 for selected student testimonials.)
- **Impact on Student Learning**: The feedback from students reflects a strong appreciation for my ability to make complex concepts understandable, my use of innovative teaching methods, and my commitment to fostering a supportive and interactive learning environment. (See Appendix T1.12.2 for examples of feedback and impact statements.)
- **Faculty Evaluation Recognition**: Rated as "Excellent" in teaching in the 2020-2022 faculty evaluation cycle at UAEU, reflecting a high level of performance and commitment to student success. (See Appendix T1.12.3 for faculty evaluation details.)

These testimonials serve as a meaningful acknowledgment of my efforts to provide high-quality education and a positive learning experience for all students.

2.13 T1.13. Grants Awarded for Teaching Innovation

While I have not received individual grants specifically for teaching innovation, I was part of a team at the Telecommunications Engineering Department at Yarmouk University that secured a €1 million grant from the Tempus Program (EU) to modernize the MSc. Program in Telecommunications Engineering. The project, titled "An Industry-Oriented MSc. Program in Telecommunications Engineering—Towards an EU Approach," focused on aligning the curriculum with European educational standards and industry needs, thereby fostering an innovative and practical approach to engineering education.

For more details on the project and my involvement, please refer to Appendix T1.13.1.

2.14 T1.14. Publication in Peer-Reviewed Pedagogical Journals of international standing

I have contributed to pedagogical research through publications in peer-reviewed journals that focus on engineering education:

- **A Master's Program in Telecommunications Management–Demand-Based Program Design,** *European Journal of Engineering Education*, Vol. 40, No. 3, pp. 267-284, 2015. (Impact factor: 2.4) - This paper presents a curriculum design methodology based on industry demand analysis, providing valuable insights into demand-based program development that enhances both student learning outcomes and industry readiness. The work contributes to the scholarship of teaching and learning in engineering education through its systematic approach to curriculum design. For supporting evidence, please refer to Appendix T1.14.1.
- **Cybersecurity Activities for Education and Curriculum Design: A Survey,** *Computers in Human Behavior Reports*, Vol. 16 (2024): 100501. (Impact factor: 5.8) - This survey examines cybersecurity education approaches and curriculum design methodologies, providing educators with frameworks for designing effective cybersecurity courses. The research emphasizes the intersection of cybersecurity and pedagogy, offering insights into educational methodologies in this critical area of study. For supporting evidence, please refer to Appendix T1.14.2.

2.15 T1.15. Presentations (oral and/or poster) at international educational conferences

- **Current Conference Engagement**: While I have not yet presented at international educational conferences, I am actively pursuing opportunities to share my pedagogical research and teaching innovations with the international education community.
- **Conference Submission Under Review**: I have submitted a paper titled "Technology-Enhanced Active Learning in Engineering Education: Promoting Equity and Engagement Through Digital Innovation" to The 7th International Conference on Modern Educational Technology (ICMET 2025), currently under review. If accepted, this presentation will share my research on the effectiveness of technology integration in promoting inclusive and engaging learning environments (see Appendix T1.15.1 for submission confirmation and paper abstract).

2.16 T1.16. Other Achievements in the Field of Teaching

- **Faculty Evaluation Recognition**: Rated as "Excellent" in teaching in the 2020-2022 faculty evaluation cycle at UAEU, reflecting a high level of performance and commitment to student success. (See Appendix T1.16.1 for faculty evaluation details.)

- **Development of Online Educational Resources**: Created a YouTube channel featuring in-class lectures from courses taught at Yarmouk University. This resource is accessible to both students and faculty, enhancing learning and teaching by providing additional support outside of traditional classroom settings. (See Appendix T1.16.2 for YouTube channel details.)
- **Implementation of Technology-Enhanced Learning**: Integrated various digital tools, such as iPad-based interactive lectures and software like MATLAB and Python, to enhance student engagement and improve learning outcomes. (See Appendix T1.16.3 for examples of technology use in teaching.)
- **Coordination of Graduation Project Groups**: Coordinated Graduation Project groups for Spring 2022, overseeing the progress of multiple student projects and providing guidance and support to ensure successful project completion. (See Appendix T1.16.4 for details on project coordination and outcomes.)
- **Participation in International Workshop**: Attended a two-week workshop at Dublin City University in 2012, focused on developing course descriptors, teaching, and assessment strategies. This workshop provided valuable insights and exposure to innovative teaching practices and resources, which I have integrated into my teaching. (See Appendix T1.16.5 for workshop details.)
- **Participation in International Workshop**: Attended a one-week workshop at Queen Mary University in 2011 focused on raising awareness about the European educational system. This workshop covered topics such as quality assurance, e-learning, distance learning, course management, and accreditation, providing valuable insights that have informed my teaching practices. (See Appendix T1.16.6 for workshop details.)
- **Participation in Industry-Focused Curriculum Development Workshop**: Attended a workshop on developing Master's programs in Telecommunications Engineering from an industrial perspective. This workshop emphasized the integration of industry standards and needs into academic curricula, enhancing the relevance and applicability of the programs. (See Appendix T1.16.7 for workshop details.)
- **Participation in Seminar on EU Educational Systems**: Attended a seminar titled "Awareness of EU Educational System" at Yarmouk University, Hijjawi College, in 2011. The seminar covered key topics such as the Bologna Process, ECTS Credits, and quality assurance in European higher education, providing insights into how these systems could inform and improve local educational practices. (See Appendix T1.16.8 for seminar details.)
- **International Teaching Exchange**: Participated in a one-week teaching exchange at the University of Deusto in Spain under the Erasmus Plus program. This experience allowed me to share my expertise in Telecommunications Engineering with an international audience and gain valuable insights into different educational practices. (See Appendix T1.16.9 for exchange program details.)
- **Continuous Improvement of Teaching Practices**: Consistently update and enhance my in-class teaching methods based on student feedback and the latest best practices in education, focusing on building upon strengths and addressing areas for improvement.
- **Faculty Teaching Academy Program (FTAP) Participation**: Completed a semester-long professional development program focused on inclusive, student-centered teaching and instructional design. Through this training, I implemented Constructivist Learning Theory principles including flipped classroom models, SMART learning objectives, and inclusive assessment strategies (see Appendix T1.6.7 for FTAP certification and implementation examples).
- **Workshop Participation on AI in Higher Education**: Attended the university-wide workshop on "Artificial Intelligence and the Future of Higher Education" in March 2025, exploring AI-enhanced pedagogy and its impact on teaching and learning. This participation demonstrates my commitment to staying current with emerging educational technologies (see Appendix T1.16.10 for workshop attendance certificate).

These achievements demonstrate my commitment to advancing teaching practices and continually improving the educational experience for students.

3 SCHOLARSHIP AND CREATIVE ACTIVITY (FORM R1)

3.1 R1.1 Peer Evaluation by External Reviewer

To be provided by the Department/College Committee in Appendix R1.1.

3.2 R1.2 Impact of Candidate's Research on His Discipline

My research portfolio comprises 45 publications in reputable venues, consisting of 27 journal publications [1]–[27], 15 conference papers [28]–[42], 2 book chapters/book publications [43], [44], and 1 doctoral dissertation [45] across diverse fields including bioinformatics, communications theory, machine learning, and wireless communications. Notably, since joining UAEU in 2020, I have contributed 20 publications to this body of work, including 17 journal articles [1]–[17] and 3 conference papers [28]–[30], demonstrating sustained research productivity and impact during my tenure at the institution. Particularly in 2024-2025, I have significantly enhanced my research profile, publishing in high-impact journals including *Ain Shams Engineering Journal* (CiteScore: 12.2, Impact Factor: 5.9), the *Journal of King Saud University -*

Engineering Sciences (CiteScore: 12.1), *IEEE Open Journal of the Communications Society* (Impact Factor: 6.1, CiteScore: 10.5), *Cluster Computing* (Impact Factor: 4.1), *IEEE Access* (Impact Factor: 3.6, CiteScore: 9.0), and *IET Smart Cities* (Impact Factor: 3.1), with improved citation metrics and h-index performance (see Appendix R1.3.1 for comprehensive list of publications, PDF files, and interactive research portfolio analysis dashboard).

My research has made significant contributions by developing novel algorithms and computational models that enhance both theoretical frameworks and practical applications. These publications reflect the broad academic impact of my work and its contribution to advancing knowledge in these areas. Through collaborations with distinguished researchers and interdisciplinary projects, my work has enriched academic discourse and paved the way for future innovations in machine learning and bioinformatics. As I continue to explore new methodologies and research avenues, I am committed to pushing the boundaries of these fields and contributing to their evolution.

Recently, my focus on machine learning has explored its potential to transform data analysis and predictive modeling, laying the groundwork for future applications in healthcare, genomics, and communication systems. This research demonstrates the versatility of machine learning and its capacity to address complex problems. Moving forward, I am committed to continuing my work in these areas, contributing innovative solutions and pushing the boundaries of these evolving fields.

3.3 R1.3 Quantity and quality of research publications, noting particularly a continuation of this effort at the UAEU

Since joining UAEU in 2020, I have maintained a robust research output, contributing to a total of **20 publications** in reputable journals and conferences. My publications span diverse fields such as bioinformatics, communications theory, machine learning, and wireless communications, reflecting my interdisciplinary approach and commitment to advancing multiple areas of study.

- **Journal Publications at UAEU (17 publications):** My recent journal publications at UAEU [1]–[17] demonstrate significant advancement in publication quality. Notable achievements include publications in high-impact journals such as the *Journal of King Saud University - Engineering Sciences* (CiteScore: 12.1), *IEEE Open Journal of the Communications Society* (Impact Factor: 6.1, CiteScore: 10.5), *Ain Shams Engineering Journal* (CiteScore: 12.2, Impact Factor: 5.9), *Cluster Computing* (Impact Factor: 4.1), *IEEE Access* (Impact Factor: 3.6, CiteScore: 9.0), *IET Smart Cities* (Impact Factor: 3.1), *Sensors* (Impact Factor: 4.42), and other reputable venues. The significant increase in publication quality, as evidenced by publications in journals with impact factors and CiteScores exceeding 10.0, demonstrates my growing research maturity and international recognition.
- **Conference Publications at UAEU (3 publications):** My conference contributions at UAEU [28]–[30] include presentations at prestigious international venues such as ICBET 2025 and the International Conference on Biomedical Engineering and Technology, demonstrating continued engagement with the international research community and dissemination of innovative research findings.
- **Submitted Manuscripts Under Review:** I currently have several manuscripts under review, including submissions to *Artificial Intelligence Review* (Springer), *Sensors* (MDPI), and *ICMET 2025* - The 7th International Conference on Modern Educational Technology. These submissions span quantum artificial intelligence, assistive technologies, and engineering education, demonstrating the breadth and interdisciplinary nature of my research portfolio.
- **Previous Research Contributions at Yarmouk University:** During my tenure at Yarmouk University, I published 14 works [18]–[27], [31]–[34] consisting of 10 journal articles and 4 conference papers across various reputable venues. My work at Yarmouk University focused on bioinformatics, communications theory, and related areas, building upon my doctoral research and establishing a strong foundation for my subsequent research at UAEU.
- **Research Contributions During Doctoral Studies at IIT:** During my PhD studies at Illinois Institute of Technology, I contributed 11 publications [35]–[45] including 8 conference papers, 2 book chapters, and my doctoral dissertation. This early research focused on applying communications theory to biological systems, particularly gene expression analysis, which laid the foundation for my interdisciplinary approach to research.
- **Research Impact and International Recognition:** My research portfolio demonstrates sustained productivity with 45 total publications consisting of 27 journal articles, 15 conference papers, 2 book chapters/book publications, and 1 doctoral dissertation [1]–[27] and [28]–[33], [35]–[45]. The evolution from foundational research during my doctoral studies at IIT to applied research at Yarmouk University to high-impact publications at UAEU reflects my commitment to advancing research quality and international visibility (see Appendix R1.3.1 for comprehensive list of publications, PDF files, and interactive research portfolio analysis dashboard).
- **Continued Research Excellence:** The publications cited above demonstrate a continued effort to advance my research in key areas such as machine learning, bioinformatics, wireless communications, and smart city technologies, while also exploring new interdisciplinary approaches. Moving forward, I am committed to further expanding the quality and impact of my research through continued publication in high-impact journals and collaborations with other leading researchers in my fields. The trajectory from my early career work to recent high-impact publications demonstrates sustained research excellence and growing international recognition in multiple domains.

3.4 R1.4 External and Internal Research Grants

I have actively secured various research grants from both external and internal sources to support my research across multiple domains. These grants have enabled me to explore innovative methodologies, contribute to interdisciplinary studies, and advance the understanding in fields such as bioinformatics, communication theory, and assistive technologies. The table below provides a summary of the grants I have been awarded, including the project titles, sources of funding, roles, budgets, and key details: Please see Appendix R1.4.1 for related documents.

Table 3: Funded Projects Secured While at UAEU

No.	Grant Period	Project Title	Source of Funding	Role	Budget	Status
1	2023—2025	Gene and Regulatory Sequence Identification using a Communications, Coding, and Information Theory Approach	Start-Up Grant, UAEU, Al Ain, UAE	Principal Investigator	367,000 AED	Completed December 2024.
2	February 2023—May 2023	Assistive Smart Glasses for the Visually Impaired	SDG Research Grant, UAEU, Al Ain, UAE	Project Supervisor	31,820 AED	Completed on April 8, 2023.
3	May 2024—May 2025	Innovative Sight-Assist Glasses: A Hybrid CNN-LSTM Approach for Enhanced Facial and Object Recognition for People with Visual Impairment	SURE Plus Grant, UAEU, Al Ain, UAE	Project Supervisor	40,000 AED	Completed May 2025.
4	May 2025—May 2026	Offline Assistive System for the Visually Impaired: Integrating Stereo Depth Perception, Object Detection, and Real-Time Feedback	SURE Plus Grant, UAEU, Al Ain, UAE	Project Supervisor	40,000 AED	Ongoing until May 2026.

Table 4: Funded Projects Secured While at Illinois Institute of Technology (IIT)

No.	Grant Period	Project Title	Source of Funding	Role	Budget	Status
1	2007—2008	Analysis of Genomic Translation Using a Communications Theory Approach	Pritzker Institute of Biomedical Science and Engineering & Illinois Institute of Technology, USA	Research Assistant	$25,000+ USD	Finished in 2008

Submitted Grant Proposals:

1. UPAR Proposal (No. 4996): "An Offline Modular Assistive System with Real-Time Stereo Vision, Gesture, and Sign Language Recognition for the Visually Impaired", 348,000 AED, 2 years (Under Review)
2. AUA–UAEU Joint Proposal (No. 4999): "Hybrid AI-DSP Framework for Enhanced Gene and Protein-Coding Region Detection", 500,000 AED, 2 years, in collaboration with National University of Singapore (NUS) (Under Review)

3.5 R1.5 Presentations at International Research Conferences

I have presented my research at 15 international and prestigious conferences [28]–[42]. Documentation of these presentations, including attendance certificates, invitation letters, and letters of gratitude, can be found in Appendix R1.5.

3.6 R1.6 Successful Supervision of Master or Doctoral Students

I have successfully supervised several Master's theses at Yarmouk University, guiding students through diverse research topics in communications and image processing. The table below provides a summary of the theses I have supervised: (Please see Appendix R1.6.1 for relevant evidence):

Table 5: Supervised Theses

No.	Student Name	Thesis Title	Institution	Year
1	Dima F. Turki	The Capacity of a Gaussian Broadcast Channel Using Relays	Yarmouk University	2019
2	Laith M. Al-Hawari	Capacity of Parallel Relay Channel with Orthogonal Components	Yarmouk University	2015
3	Monther Alrawashdeh	Capacity of Cascade Relay Channel with Orthogonal Components	Yarmouk University	2015
4	Bassam M. Alrawashdeh	Canny Edge Detection Enhancement by Scale Multiplication	Yarmouk University	2013
5	Ali T. Alzawahreh	Enhanced Image Segmentation Algorithms	Yarmouk University	2013

Current PhD Committee Participation at UAEU: I am actively serving as a member of PhD advisory committees, contributing to doctoral student guidance and evaluation:

1. Member, PhD Advisory Committee for Rola Abdallah – Electrical Engineering: Communications (Ongoing)
2. Member, PhD Prospectus Exam Committee for Hanan Abdelmotagally – AI and ML in Cyber-Physical Systems (May 2025)

(See Appendix R1.6.2 for committee appointment letters and participation records.)

3.7 R1.7 Other Achievements in the Field of Scholarship

- **Faculty Evaluation Recognition**: I was rated "Excellent" in research during the faculty evaluation cycle at UAEU from August 2020 to May 2022, reflecting my strong commitment and substantial contributions to scholarly activities. The evaluation process at UAEU is thorough, assessing faculty performance across teaching, research, and service. It involves multiple levels of review, beginning with a departmental committee, followed by the department chair, and concluding with a final evaluation by the Dean of the College of Engineering. (Refer to Appendix R1.7.1 for detailed evaluation evidence.)
- **Session Chair at International Conferences:** I served as a session chair for "Medical Imaging and Biomedical Image Processing" at the 2023 13th International Conference on Biomedical Engineering and Technology (ICBET 2023), held from June 15-18, 2023, in Tokyo, Japan. This role involved moderating discussions, facilitating Q&A sessions, and ensuring smooth transitions between presentations. Additionally, I was invited to chair a session at the IWBBIO 2014 (International Work-Conference on Bioinformatics and Biomedical Engineering), held from April 7-9, 2014, in Granada, Spain. My responsibilities included overseeing the session's presentations and providing expert evaluations to help select contributions for special issues in relevant journals. (See Appendix R1.7.2 for the Session Chair Certificate and related correspondence.)
- **Technical Committee Membership at International Conference**: I serve as a member of the Technical Committee for the International Conference on Biomedical Engineering and Technology (ICBET). As a committee member, I contribute to reviewing conference submissions, providing expert opinions on research quality, and shaping the direction of the conference's scientific program. (See Appendix R1.7.3 for the Session Chair Certificate and related correspondence.)
- **Editorial and Review Service**: I serve as a reviewer for multiple high-impact journals including PLOS ONE, IEEE Transactions on Microwave Theory and Techniques, and the Emirates Journal for Engineering Research. Additionally, I served as a reviewer for the 10th UAE Graduate Students Research Conference (GSRC 2025), contributing to the evaluation of emerging research in the region (see Appendix R1.7.4 for reviewer invitations and acknowledgments).
- **Organizing Committee Leadership**: I serve as Organizing Committee Administrator for IEEE BioCAS 2025 and as a Technical Program Committee Member for IEEE AICT 2025 – 19th International Conference on Application of ICT, Al Ain, UAE (October 29–31, 2025). These roles demonstrate my growing leadership in the international research community and my commitment to advancing scholarly conferences in my field (see Appendix R1.7.5 for appointment letters and committee responsibilities).

4 UAEU AND COMMUNITY SERVICE (FORM S1)

4.1 S1.1. Service in Academic Management

1. **Coordinator Roles**:
   - **Graduation Project Coordinator (GPI, ELEC 585)**: Coordinated the Graduation Project I course for Electrical and Communication Engineering students during Fall 2021 and Spring 2022. Responsibilities included organizing project assignments, managing deadlines, and providing guidance to ensure the successful completion of student projects. (See Appendix S1.1.1 for coordination assignment details and course management documentation.)
   - **Lab Coordinator**: Served as the coordinator for the "Communications and Microprocessors Lab." (See Appendix S1.1.2 for supporting documents.)

- **Chair, Department Travel Committee (AY 2025/2026)**: Currently serving as Chair of the Department Travel Committee, overseeing the review and approval of faculty travel requests while ensuring compliance with university policies and budget allocations (see Appendix S1.1.12 for committee appointment and responsibilities).

2. **Departmental and Program Coordination**:
   - **Academic Advisor for ZMC Communication Engineering Students – Batch 45**: Managed the academic progress of a specific cohort of communication engineering students, providing mentorship, monitoring academic performance, and ensuring students meet their educational objectives. (See Appendix S1.1.3 for supporting documents.)
   - **Secretary for the Electrical and Communication Engineering Department Council (AY 2021-2022)**: Responsible for documenting meeting minutes, managing departmental communications, and coordinating agendas for council meetings. (See Appendix S1.1.5 for meeting minutes.)
   - **Digital Liaison for Educational Technology Integration**: Appointed as Digital Liaison for the College of Engineering (Spring 2025), supporting faculty in adopting emerging educational technologies and best practices in digital learning through coordination with CETL's course transformation initiatives (see Appendix S1.1.13 for liaison appointment and project outcomes).
3. **Committee Membership and Contributions**:
   - **ECE Department Assessment Committee**: Participated in the evaluation of departmental programs and curricula, contributing to the continuous improvement of academic standards and learning outcomes. (See Appendix S1.1.5 for committee meeting notes and assessment reports.)
   - **Smart Learning Operational Committee (SLOC)**: Represented the department in this university-level committee, contributing to the development and implementation of smart learning initiatives across the campus. (See Appendix S1.1.6 for committee correspondence and project summaries.)
   - **Faculty/Instructor Performance Evaluation Committee (AY 2024/2025)**: Served as a member of the departmental committee responsible for evaluating faculty and instructor performance, contributing to personnel decisions and professional development recommendations (see Appendix S1.1.14 for committee assignments and evaluation protocols).
   - **Grade Appeal Committee**: Appointed by the Dean to serve on Grade Appeal Committees for ELEC 325 and ECOM 360, reviewing student grade disputes and ensuring fair academic evaluation processes (May 2025) (see Appendix S1.1.15 for committee appointments and case summaries).
   - **Fact-Finding Committee on Academic Integrity**: Served as a member of the Fact-Finding Committee on Academic Integrity for the College of Engineering (December 2024), investigating academic integrity violations and recommending appropriate actions (see Appendix S1.1.16 for committee appointment and confidential proceedings summary).
4. **Program and Curriculum Development**:
   - **Biomedical Engineering and Informatics Program Development Workgroup**: Actively engaged in the development of the new program, focusing on curriculum design and program planning to align with industry needs and academic standards. (See Appendix S1.1.7 for program development notes and curriculum drafts.)
   - **Course Revision for Communications Engineering Program**: Contributed as a member of the committee responsible for updating and revising the communications engineering curriculum to incorporate modern practices and technologies. (See Appendix S1.1.8 for course revision proposals and supporting documents.)
5. **Departmental Leadership Role at Yarmouk University**: I served as the Chair of the Telecommunications Engineering Department at Yarmouk University for two years, from 2016 to 2018. In this role, I managed departmental operations, provided leadership to faculty and staff, and implemented strategic initiatives to improve the department's academic and research performance. (See Appendix S1.1.9 for relevant documentation.)
6. **Chair of ABET Accreditation Committee at Yarmouk University**: I served as the Chair of the ABET Committee at Yarmouk University, leading efforts to achieve ABET accreditation for the BSc in Telecommunications Engineering program. In this role, I oversaw the accreditation process, ensuring that the program met the rigorous standards required for accreditation. (See Appendix S1.1.10 for relevant documentation.)
7. **Plenary Session Speaker, HTU 2018 Conference:** I spoke at **HTU 2018**, participating in **Plenary Session III: Youth & Future Technologies**, discussing advancements in technology and its impact on youth. As **Chairman of Telecommunications Engineering at Yarmouk University**, I shared insights on future technological trends. (See Appendix S1.1.11 for evidence.)

4.2 S1.2. Participation in the Activities of National, Regional, International Professional Organization/Associations Committee

- **Technical Committee Member, ICBET:** Served as a technical committee member for the International Conference on Biomedical Engineering and Technology (ICBET), contributing to paper reviews and conference planning. (See Appendix S1.2.1 for evidence.)

- **Session Chair, ICBET 2023:** Chaired a session on "Medical Imaging and Biomedical Image Processing" at ICBET 2023. (See Appendix S1.2.2 for the Session Chair Certificate.)
- **Session Chair, IWBBIO 2014**: Chaired a session on "Bioinformatics and Biomedical Image Processing" at IWBBIO 2014 in Granada, Spain. (See Appendix S1.2.3 for documentation.)
- **Tempus Program Participant:** Contributed to the modernization of the MSc in Telecommunications Engineering at Yarmouk University through a 1 million Euro EU Tempus grant. (See Appendix S1.2.4 for details.)
- **IEEE Communications Society Member:** Active member of the IEEE Communications Society, staying engaged in relevant academic and industrial developments. (See Appendix S1.2.5 for membership evidence.)
- **Member of Arab American Association of Engineers and Architects (AAAEA):** Participated in the activities and events of the AAAEA, contributing to the broader engineering community. (See Appendix S1.2.6 for membership details.)
- **Member of Jordan Engineers Association:** Active member since 2002, involved in various initiatives to advance engineering in Jordan and the region.
- **Organizing Committee Member, MECBME 2020**: I served as a member of the Organizing Committee for the Middle East Conference on Biomedical Engineering (MECBME) 2020. (See Appendix S1.2.7 for evidence of participation.)
- **Judge, IEEE UAE Student Day**: Served as a judge at IEEE UAE Student Day, evaluating student projects and providing constructive feedback on their designs. (See Appendix S1.2.10 for judging event details.)
- **IEEE BioCAS 2025 Organizing Committee Administrator**: Serving as Organizing Committee Administrator for IEEE BioCAS 2025, responsible for coordinating organizational aspects of this major international conference in biomedical circuits and systems (see Appendix S1.2.8 for appointment letter and responsibilities).
- **IEEE AICT 2025 Technical Program Committee Member**: Serving as Technical Program Committee Member for IEEE AICT 2025 – 19th International Conference on Application of ICT, Al Ain, UAE (October 29–31, 2025), contributing to the technical review process and program development (see Appendix S1.2.9 for TPC appointment and review assignments).

4.3 S1.3. Provision of Consultancy or Advisory Services

- **Member of UAEU's Smart Learning Operational Committee (SLOC):** Member of the **Smart Learning Operational Committee** at UAEU. (See Appendix S1.3.1 for supporting documents.)
- **Member of UAEU's Innovation Task Force Committee:** Member of the **Innovation Task Force Committee** at UAEU. (See Appendix S1.3.2 for evidence.)
- **Development of Biomedical Engineering and Informatics Program:** Contributed to the curriculum design and strategic development of UAEU's **Biomedical Engineering and Informatics Program**, aligning it with industry needs. (See Appendix S1.3.3 for program documentation.)
- **Reviewer for the Emirates Journal for Engineering Research:** Act as a reviewer for the **Emirates Journal for Engineering Research**, providing expert feedback on submitted articles in the field of engineering. (See Appendix S1.3.4 for reviewer acknowledgments.)
- **Technical Program Committee (TPC) Member and Reviewer for ICBET Conference:** Serve as a **TPC member and reviewer** for the **ICBET Conference**. (See Appendix S1.3.5 for evidence.)
- **Judge for Ibn Trais Educational Award, Emirates Falcon International Private School (May 2024):** Evaluated student innovative projects and presentations, provided feedback, and contributed to the selection of award recipients. (See Appendix S1.3.6 for acknowledgment.)
- **Tempus Project Consultancy (2015-2018):** Led the **demand and competition analysis** for the modernization of Yarmouk University's MSc. Telecommunications Engineering program under the EU-funded **Tempus Project**. (See Appendix S1.3.7 for project details.)
- **Judge, Ibn Trais Educational Award (2025)**: Served as a judge for the Ibn Trais Educational Award, evaluating innovative student projects and presentations while providing constructive feedback to young innovators (see Appendix S1.3.8 for judging appointment and evaluation criteria).
- **Reviewer for Multiple International Journals**: Actively serving as a reviewer for several high-impact international journals including PLOS ONE, and IEEE Transactions on Microwave Theory and Techniques, contributing to the peer review process and maintaining publication standards (see Appendix S1.3.9 for reviewer invitations and acknowledgments).

4.4 S1.4. Contribution to Planning/Delivering of Development Activities

- **Exponential Thinking Talks**: I delivered several talks on "Exponential Thinking" at Yarmouk University and Hashemite University, which resonated with faculty and staff, eventually influencing the proposal of an Online Education Center at Yarmouk University. The talks are recorded on my YouTube channel and are accessible to faculty at UAEU, providing ongoing professional development in this area. (See Appendix S1.4.1 for evidence.)

- **Biomedical Engineering Program Development**: I actively contributed to the curriculum planning and development of the Biomedical Engineering and Informatics Program at UAEU. This project, aligned with industry trends, has served as a framework for faculty to stay updated on current developments in the field. (See Appendix S1.4.2 for evidence.)
- **Proposal for Online Education Center**: Before joining UAEU, I developed a proposal for an Online Education Center at Yarmouk University, which was a direct result of my work on digital and exponential thinking education, aimed at enhancing faculty engagement in online teaching and learning strategies. (See Appendix S1.4.3 for evidence.)
- **Session Chair at ICBET and IWBBIO Conferences**: I served as a session chair at multiple international conferences, such as ICBET and IWBBIO, where I facilitated discussions that contributed to faculty development and collaborative learning in the fields of bioinformatics and engineering. (See Appendix S1.4.4 for evidence.)
- **Innovation in Pedagogy:** I continuously share my experience and effective teaching practices with colleagues in the department, especially regarding lecture delivery techniques. For instance, I always record my lectures and make them available to students through **Blackboard** or **MS Teams**, ensuring flexible access to course material. Additionally, I create an MS Teams class for each course I teach, which facilitates communication and resource sharing. By using an **iPad** to deliver my lectures, I can integrate various applications during class, promoting a highly interactive learning environment. This approach has been well-received by both students and faculty, leading to its adoption by other instructors in the department. (See Appendix S1.4.5 for evidence.)
- **Organizer of Exponential Day at Yarmouk University (2016-2018):** Organized the Exponential Day event, featuring expert talks on exponential growth in various technologies, including AI and 5G networks. The event's content, shared on my YouTube channel, also inspired the proposal for an **Online Education Center** at YU. (See Appendix S1.4.6 for evidence.)
- **Innovation Competition Organization**: I organized and hosted the INNOSPARK Award 2025, an innovation-focused student competition held on February 4, 2025. This event promoted student innovation, presentation skills, and teamwork while providing mentorship and networking opportunities. I coordinated logistics, facilitated sessions, and moderated key discussions to enhance students' creative and entrepreneurial capabilities (see Appendix S1.4.7 for event documentation and participant feedback).
- **Faculty Teaching Academy Program (FTAP) Participation**: Completed comprehensive training in Constructivist Learning Theory (CLT) principles, implementing evidence-based approaches including flipped classroom models, SMART learning objectives, and inclusive assessment strategies. This training strengthened my ability to design teaching processes that achieve course outcomes while promoting equity and meeting diverse learning needs (see Appendix S1.4.8 for FTAP certification and training materials).

4.5 S1.5 Contribution and Commitment to the Application of International Standards

- **Chair of ABET Accreditation Committee at Yarmouk University (2016-2018):** Led efforts to secure ABET accreditation for the Telecommunications Engineering program, ensuring the curriculum met internationally recognized quality standards. (See Appendix S1.5.1 for evidence.)
- **Tempus Project Workgroup Leader (2015-2018):** Contributed to modernizing Yarmouk University's MSc Telecommunications Engineering program under the EU-funded Tempus Project, aligning it with European Union standards for higher education. (See Appendix S1.5.2 for project details.)
- **Ensuring International Standards in Graduation Projects:** I ensure all of my Graduation Project students apply international standards in their work, fostering high-quality outcomes that align with global practices in engineering and technology. (See Appendix S1.5.3 for supporting evidence.)
- **Member of the ECE Assessment Committee (2021-present):** As a member of the Electrical and Communications Engineering assessment committee, I help ensure that our academic programs consistently meet international standards for quality and effectiveness. (See Appendix S1.5.4 for relevant documentation.)
- **Quality Matters (QM) Certification Training**: I am scheduled to complete the Quality Matters "Applying the QM Rubric" certification training on September 23, 2025, sponsored by CETL, which represents a commitment to implementing internationally recognized quality assurance standards in course design and delivery (see Appendix S1.5.5 for QM training enrollment and curriculum details).
- **Programs Update Committee Participation**: Active participation in multiple Programs Update Committees during AY 2024-2025 has ensured that curricular reviews and program updates align with international accreditation standards and best practices in engineering education (see Appendix S1.5.6 for committee reports and standards compliance documentation).
- **International Conference Standards and Best Practices**: Through Session Chair roles at ICBET and IWBBIO conferences, and current leadership positions in IEEE BioCAS 2025 and IEEE AICT 2025, I actively contribute to maintaining international standards in research presentation and conference organization. (See Appendix R1.7.2, Appendix S1.2.8, and Appendix S1.2.9 for conference leadership documentation.)

4.6 S1.6 Participation in Peer Evaluations for Academic Purposes

- **Comprehensive Exam Participation (ECE Department, UAEU)**: Served on comprehensive exam committees for MS and PhD students, contributing to the design and evaluation of exams. I have also chaired one of these committees. (See Appendix S1.6.1 for records.)
- **Internal Examiner for Master's Thesis Defense (UAEU)**: Acted as an internal examiner, providing detailed feedback on master's thesis defenses. (See Appendix S1.6.2 for records.)
- **Reviewer and TPC Member for ICBET Conference**: Reviewed papers and served on the Technical Program Committee for ICBET. (See Appendix S1.6.3 for evidence.)
- **Session Chair for ICBET and IWBBIO Conferences**: Chaired sessions at ICBET and IWBBIO, ensuring quality presentations and discussions. (See Appendix S1.6.4 for certificates.)
- **Reviewer for the Emirates Journal for Engineering Research**: Contributed to the evaluation of research articles in the field of engineering as a reviewer for the *Emirates Journal for Engineering Research*. (See Appendix S1.6.5 for reviewer acknowledgment.)
- **Examination Committee Participation for Graduation Projects (Chair/Department Representative)**: Acted as a chair or department representative on several graduation project examination committees, contributing to the evaluation and final assessment of student projects. (See Appendix S1.6.6 for records.)
- **PhD Committee Participation**: Serving on multiple PhD committees, including as a member of the PhD Advisory Committee for Rola Abdallah in Electrical Engineering: Communications and as a member of the PhD Prospectus Exam Committee for Hanan Abdelmotagally in AI and ML in Cyber-Physical Systems (May 2025). These roles involve evaluating research proposals, providing guidance on dissertation development, and ensuring academic rigor (see Appendix S1.6.7 for committee appointment letters and evaluation records).
- **Reviewer for 10th UAE Graduate Students Research Conference (GSRC 2025)**: Served as a reviewer for the 10th UAE Graduate Students Research Conference, evaluating research submissions from graduate students across the UAE and providing constructive feedback to support emerging researchers (see Appendix S1.6.8 for reviewer invitation and evaluation summaries).

4.7 S1.7 Contribution to the Student's Advising and Counselling Activities, Extra-curricular Activities or Any Other Activity Pertaining to Student Services

- **Academic Advisor (Batch 45)**: Managed the academic progress and mentoring for over 39 communication engineering students. (See Appendix S1.7.1 for advising records.)
- **Higher Education Support and Career Counseling**: Supported students in pursuing higher education and provided reference letters for their applications. (See Appendix S1.7.2 for reference letters.)
- **Supervision of Senior Projects**: Guided senior projects (GPI and GPII), including smart glasses and home automation systems. I provided project management guidance, mentoring, and technical expertise to ensure the projects' success. (See Appendix S1.7.3 for project details.)
- **Judge at IEEE UAE Student Day and Ibn Trais Award**: Evaluated student projects, offering feedback on their designs. (See Appendix S1.7.4 for judging event details.)
- **Orientation Day 2022 Participation**: Presented to new Electrical and Communication Engineering students, helping them transition into the program. (See Appendix S1.7.5 for presentation materials.)
- **Career Guidance for Graduating Students**: Advised graduating students on career paths and job opportunities. (See Appendix S1.7.6 for counseling evidence.)
- **Supervision of Industrial Training**: Supervised students during their industrial training, providing guidance, feedback, and ensuring they meet their learning objectives. (See Appendix S1.7.7 for supervision details and reports.)
- **Coordination of Graduation Projects**: Oversaw Spring 2022 Graduation Projects, guiding student groups and ensuring successful project outcomes. (See Appendix S1.7.8 for project coordination details.)
- **INNOSPARK Award 2025 Organization and Student Mentorship**: Organized and hosted the INNOSPARK Award 2025 on February 4, 2025, providing comprehensive mentorship to participating students. This innovation competition enhanced students' presentation skills, teamwork abilities, and entrepreneurial thinking while offering valuable networking opportunities with industry professionals and faculty (see Appendix S1.7.9 for event documentation and student feedback).
- **SURE Plus Grant Student Supervision**: Currently supervising undergraduate research students under the SURE Plus Grant (G00005245) for the "Offline Assistive System for the Visually Impaired: Integrating Stereo Depth Perception, Object Detection, and Real-Time Feedback" project, providing hands-on research experience and mentorship in computer vision and assistive technologies (see Appendix S1.7.10 for supervision records and student progress reports).

4.8 S1.8 Contribution to Development of Orientation Program

- **Orientation Day Presentation for Electrical Engineering Students (2022):** Delivered an informative presentation for new Electrical Engineering students during Orientation Day 2022. The session helped students transition

smoothly into the program and introduced them to department activities and expectations. (See Appendix S1.8.1 for presentation and certificate details).

- **Academic Advisor for Batch 45:** Provided academic advising and mentorship to over 39 communication engineering students, helping guide their academic progress and offering personalized support for their educational and professional development. (See Appendix S1.8.2 for evidence).

4.9 S1.9 Organization of Professional Workshop and/or Training Off-Campus

- **Exponential Day Event Organization (YU)**: Organized the "Exponential Day" event at Yarmouk University in collaboration with external speakers, which involved talks on exponential thinking and disruptive innovations. The event also hosted discussions on future technologies and their impact on youth. (See Appendix S1.9.1 for event agenda and outcomes.)
- **Tempus Project Workshops (2015-2018)**: Under the Tempus Project, I contributed to the organization of workshops aimed at modernizing the MSc Telecommunications Engineering program at Yarmouk University. These workshops involved both faculty and external industry stakeholders. (See Appendix S1.9.2 for workshop details and outcomes.)
- **Exponential Thinking Workshops (Multiple Universities)**: Delivered a series of talks on "Exponential Thinking and Disruptive Innovations" at various universities, including Yarmouk University and Hashemite University. These talks focused on the impact of exponential technologies and motivated the establishment of an Online Education Center at Yarmouk University. (See Appendix S1.9.3 for workshop recordings and evidence.)
- **Plenary Session Speaker, HTU 2018 Conference**: Presented at the HTU 2018 Conference in Plenary Session III: Youth & Future Technologies, discussing advancements in technology and their impact on future generations. As the Chairman of the Telecommunications Engineering Department at Yarmouk University, I contributed insights on technological trends, influencing discussions at the conference. (See Appendix S1.9.4 for evidence.)

4.10 S1.10 Membership of Editorial/Advisory Boards of Scientific Journals

- **Reviewer for Emirates Journal for Engineering Research**: Provided expert evaluations on submitted articles, ensuring the journal's adherence to high academic standards. (See Appendix S1.10.1 for reviewer invitations and acknowledgments.)
- **Technical Program Committee (TPC) Member for ICBET Conference**: Reviewed and selected papers for the ICBET Conference, contributing to the quality of the technical program. (See Appendix S1.10.2 for TPC evidence.)
- **Reviewer for IEEE Access**: Offered detailed reviews for IEEE Access submissions, contributing to maintaining the publication's technical quality. (See Appendix S1.10.3 for reviewer invitations.)
- **Reviewer for Biomedical Signal Processing and Control (BSPC) Journal**: Reviewed articles for BSPC, ensuring the journal's contribution to signal processing and biomedical research. (See Appendix S1.10.4 for reviewer invitations.)
- **Expanded Journal Review Portfolio**: Extended my reviewing activities to include multiple high-impact journals: PLOS ONE, and IEEE Transactions on Microwave Theory and Techniques, demonstrating recognition of my expertise across diverse research domains (see Appendix S1.10.5 for additional reviewer invitations and acknowledgments).

4.11 S1.11 Refereeing research papers submitted for publication in scientific periodicals or conference proceedings

- **Reviewer for Emirates Journal for Engineering Research**: Evaluated multiple research papers, offering constructive feedback for publication decisions. (See Appendix S1.11.1 for evidence.)
- **Reviewer for IEEE Access**: Reviewed research submissions in engineering and technology. (See Appendix S1.11.2 for invitations.)
- **Reviewer for Biomedical Signal Processing and Control (BSPC) Journal**: Provided critical evaluations for submitted papers on signal processing. (See Appendix S1.11.3 for invitations.)
- **Reviewer for ICBET Conference**: Reviewed conference papers in the field of biomedical engineering and technology. (See Appendix S1.11.4 for evidence.)
- **Comprehensive Peer Review Activities**: Significantly expanded my peer review contributions across multiple venues, including PLOS ONE, IEEE Transactions on Microwave Theory and Techniques, and the 10th UAE Graduate Students Research Conference (GSRC 2025). This extensive review activity demonstrates my commitment to maintaining high standards in academic publishing and supporting the research community (see Appendix S1.11.5 for comprehensive review portfolio and editor acknowledgments).

4.12 S1.12 Other Achievements in the Field of Service

- **YouTube Channel: Educational Outreach and Community Engagement**: Created and maintained the "Dr. Bataineh" YouTube channel, which features over 411 videos covering topics like mathematics, engineering, and exponential thinking. The channel has reached 10.8K subscribers and is a valuable resource for both students and the

wider community. It also serves as a platform for delivering online educational content, including recorded lectures, tutorials, and professional talks on innovation and technology. (See Appendix S1.12.1 for channel link and analytics.)

- **Sharing Teaching Practices**: Shared course materials and methods with new faculty, enhancing teaching in the department. (See Appendix T1.11.1.)
- **Biomedical Engineering Program**: Helped design the Biomedical Engineering Program, aligning it with industry standards. (See Appendix T1.7.1.)
- **Teaching Exchange**: Participated in an Erasmus Plus exchange in Spain, improving teaching at UAEU. (See Appendix T1.16.9.)
- **Innovation Competition Leadership**: Successfully organized and executed the INNOSPARK Award 2025, demonstrating leadership in fostering student innovation and entrepreneurship. This initiative has become a model for similar competitions within the College of Engineering, promoting creative problem-solving and collaboration among students (see Appendix S1.12.2 for competition impact assessment and institutional recognition).
- **Professional Development Leadership**: Through my roles as Digital Liaison and FTAP participant, I have contributed to advancing teaching and learning practices across the College of Engineering, sharing innovative pedagogical approaches and supporting colleagues in adopting technology-enhanced instruction methods (see Appendix S1.12.3 for professional development contributions and colleague feedback).

Publications

[1] Z. Albataineh, H. B. Salameh, M. Al Bataineh, and A. M. Alshorman, "Adaptive beam pairing and local interpolation for robust analytical beam training in RIS-assisted wideband THz systems," *IEEE Open Journal of the Communications Society*, 2025.

[2] M. H. Alrashdan, Z. Al-qudah, and M. Al Bataineh, "Enhancing impedance control in microstrip lines using the Taguchi approach," *Journal of King Saud University–Engineering Sciences*, vol. 37, no. 5, p. 24, 2025.

[3] K. Khurshid, K. Khurshid, M. U. Hadi, M. Al Bataineh, and N. Saeed, "Securing AIoT surveillance: Techniques, challenges, and solutions," *IEEE Open Journal of the Communications Society*, 2025.

[4] M. Al Bataineh, D. I. A. Abdoun, and M. Al Ahmad, "Comparative analysis of machine learning algorithms for antenna alignments," *IEEE Access*, 2025.

[5] Z. Albataineh, H. B. Salameh, and M. Al Bataineh, "Two-stage power allocation for IRS-assisted downlink NOMA: Intra-group optimization and inter-group allocation," *IEEE Access*, 2025.

[6] Z. Albataineh, M. Al Bataineh, K. F. Hayajneh, and R. Al Athamneh, "Advanced 5G channel estimation in mmWave MIMO systems: Leveraging compressive sensing for enhanced performance," *IEEE Access*, 2025.

[7] A. Alhawatmeh, M. Al Bataineh, N. Alashqar, and R. Khalil, "Atomic solutions to Bateman–Burgers type equation via tensor products," *Partial Differential Equations in Applied Mathematics*, vol. 13, p. 101 102, 2025.

[8] N. Jayachandran, A. Abdrabou, M. Al Bataineh, and K. A. Noordin, "An architecture and realization of a smart city digital twin with eHealth case studies," *IET Smart Cities*, 2025.

[9] Z. Albataineh, K. Hayajneh, H. Shakhatreh, R. Al Athamneh, and M. Al Bataineh, "Adaptive machine learning-enhanced channel estimation for RIS-assisted mmWave systems: A hybrid approach," *Cluster Computing*, 2025.

[10] M. Ismail, N. T. Madathil, M. Alalawi, *et al.*, "Cybersecurity activities for education and curriculum design: A survey," *Computers in Human Behavior Reports*, vol. 16, p. 100 501, 2024.

[11] M. H. Alrashdan, Z. Al-qudah, and M. Al Bataineh, "Microstrip patch antenna directivity optimization via Taguchi method," *Ain Shams Engineering Journal*, vol. 15, no. 9, p. 102 923, 2024.

[12] S. Al-Shihabi, R. Aydin, Z. C. Araci, F. Dweiri, M. Obeidat, and M. F. Al Bataineh, "Abolishing single-use plastic water bottles in Dubai hotels as a voluntary act—scenarios and environmental impacts," *Sustainability*, vol. 16, no. 8, p. 3121, 2024.

[13] M. Al Bataineh, D. I. A. Abdoun, H. Alnuaimi, Z. Al-Qudah, Z. Albataineh, and M. Al Ahmad, "Head impact detection using machine learning algorithms," *IEEE Access*, vol. 12, pp. 4938–4947, 2024.

[14] M. Al Bataineh, M. M. Umar, A. Moin, M. I. Hussein, and M. Al Ahmad, "Classification and prediction of communication cables length based on S-parameters using a machine-learning method," *IEEE Access*, vol. 11, pp. 108 041–108 049, 2023.

[15] R. Shams, A. Abdrabou, M. Al Bataineh, and K. A. Noordin, "Managing energy consumption of devices with multiconnectivity by deep learning and software-defined networking," *Sensors*, vol. 23, no. 18, p. 7699, 2023.

[16] M. Al Bataineh, Z. Al-qudah, A. Abdrabou, and A. N. Sandokah, "Geometric probability analysis of meeting probability and intersection duration for triple event concurrency," *Mathematics*, vol. 11, no. 12, p. 2708, 2023.

[17] Z. Al-qudah, M. Al Bataineh, and A. Musa, "A novel multiple access diamond channel model," *International Journal of Communication Systems*, vol. 33, no. 17, e4616, 2020.

[18] Z. Al-Qudah, L. Al-Hawary, M. Alrwashdeh, and M. Al Bataineh, "Parallel relay network with orthogonal components: Capacity and power allocation," *Wireless Personal Communications*, vol. 96, no. 3, pp. 4869–4884, 2017.

[19] Z. Al-qudah and M. Al Bataineh, "Allocation of resources for the Gaussian multiple access channel with practical partial cooperation," *AEU-International Journal of Electronics and Communications*, vol. 74, pp. 145–149, 2017.

[20] M. Al Bataineh and Z. Al-qudah, "A novel gene identification algorithm with Bayesian classification," *Biomedical Signal Processing and Control*, vol. 31, pp. 6–15, 2017.

[21] M. Al Bataineh, Z. Al-qudah, and A. Al-Zaben, "Iterative sequential Monte Carlo algorithm for motif discovery," *IET Signal Processing*, vol. 10, no. 5, pp. 504–513, 2016.

[22] Z. Al-Qudah, M. Alrwashdeh, L. Al-Hawary, and M. Al Bataineh, "On the capacity of a relay network with orthogonal components," *AEU-International Journal of Electronics and Communications*, vol. 70, no. 5, pp. 636–642, 2016.

[23] A. Al-Zaben, M. Al Bataineh, and S. Al-Refaie, "Temperature compensation of fiber Bragg gratings manometry catheter using Kalman filter," *Int. J. Pharm. Med. Biol. Sci*, vol. 5, pp. 12–16, 2016.

[24] K. M. Gharaibeh, H. Kaylani, N. Murphy, *et al.*, "A masters programme in telecommunications management–demand-based curriculum design," *European Journal of Engineering Education*, vol. 40, no. 3, pp. 267–284, 2015.

[25] Z. Al-qudah and M. Al Bataineh, "Cognitive interference channel: Achievable rate region and power allocation," *IET Communications*, vol. 9, no. 2, pp. 249–257, 2015.

[26] Z. Al-qudah, M. Al Bataineh, and W. Abu Shehab, "On optimal power allocation for Gaussian broadcast channel," *Innovative Systems Design and Engineering*, vol. 7, no. 6, pp. 1–7, 2016, ISSN: 2222-1727. [Online]. Available: http://www.iiste.org/Journals/index.php/ISDE/article/view/31365.

[27] M. F. Al Bataineh and Z. J. Al-Qudah, "Bayesian classification of ribosome binding sites in prokaryotic genome sequences: A communications theory approach," *International Journal of Bioscience, Biochemistry and Bioinformatics*, vol. 7, no. 3, pp. 133–142, Jul. 2017. DOI: 10.17706/ijbbb.2017.7.3.133-142.

[28] M. Al Bataineh, D. I. Abu-Abdoun, S. Al-Shihabi, and K. Muhammad, "A comparative analysis of computational approaches for gene detection," in *ICBET 2025*, 2025.

[29] M. F. Al Bataineh, "Enhanced detection and localization of zinc finger proteins using advanced neural network techniques," in *Proceedings of the 2024 14th International Conference on Biomedical Engineering and Technology*, 2024, pp. 1–4.

[30] M. F. Al Bataineh, "Communications theory-inspired algorithms for detecting protein-coding regions in prokaryotic genomes: A comparative study: Using communications theory for gene detection in prokaryotic genomes: A comparative analysis of correlation-based algorithms and Bayesian classifiers," in *Proceedings of the 2023 13th International Conference on Biomedical Engineering and Technology*, 2023, pp. 51–55.

[31] M. Al Bataineh, "Identification of coding regions in prokaryotic DNA sequences using Bayesian classification," in *International Work-Conference on Bioinformatics and Biomedical Engineering*, Springer, 2020, pp. 3–14.

[32] M. Al Bataineh, L. Huang, and G. E. Atkin, "Transcription factor binding site detection algorithm using distance metrics based on a position frequency matrix concept," in *IWBBIO*, 2014, pp. 715–726.

[33] M. Al Bataineh, L. Huang, and G. Atkin, "TFBS detection algorithm using distance metrics based on center of mass and polyphase mapping," in *2012 7th International Symposium on Health Informatics and Bioinformatics*, IEEE, 2012, pp. 37–40.

[34] A. Al-Zaben, M. Al Bataineh, and S. Al-Refaie, "Temperature compensation of fiber bragg gratings manometry catheter using kalman filter," in *Proceedings of the 2016 International Conference on Biomedical Engineering and Technology*, Singapore, 2016.

[35] L. Huang, M. Al Bataineh, A. F. Acedo, G. Atkin, X. Deng, and W. Zhang, "Identification of transcriptional promoter sequence based on statistical filter bank model," in *2010 IEEE International Conference on Electro/Information Technology*, IEEE, 2010, pp. 1–5.

[36] M. Al Bataineh, L. Huang, A. F. Acedo, G. Atkin, and N. Menhart, "An augmented block code model for protein translation using free energy based distance decoders," in *2010 IEEE International Conference on Electro/Information Technology*, IEEE, 2010, pp. 1–6.

[37] L. Huang, M. Al Bataineh, G. E. Atkin, *et al.*, "Identification of transcription factor binding sites based on the chi-square (χ^2) distance of a probabilistic vector model," in *2009 International Conference on Future BioMedical Information Engineering (FBIE)*, 2009, pp. 73–76. DOI: 10.1109/FBIE.2009.5405793.

[38] M. A. Bataineh, M. Alonso, L. Huang, G. E. Atkin, and N. Menhart, "Effect of mutations on the detection of translational signals based on a communications theory approach," in *2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2009, pp. 3853–3856. DOI: 10.1109/IEMBS.2009.5332573.

[39] L. Huang, M. A. Bataineh, G. E. Atkin, S. Wang, and W. Zhang, "A novel gene detection method based on period-3 property," in *2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2009, pp. 3857–3860. DOI: 10.1109/IEMBS.2009.5332576.

[40] M. Al Bataineh, L. Huang, I. Muhamed, N. Menhart, and G. E. Atkin, "Gene expression analysis using communications, coding and information theory based models," in *Proceedings of the 2009 International Conference on Bioinformatics & Computational Biology (BIOCOMP)*, Jul. 2009, pp. 181–185.

[41] M. Al Bataineh, M. Alonso, G. E. Atkin, S. Wang, and W. Zhang, "An optimized ribosome binding model using communication theory concepts," in *Proceedings of the 2007 International Conference on Bioinformatics & Computational Biology (BIOCOMP)*, Jun. 2007, pp. 345–348.

[42] M. Al Bataineh, M. Alonso, S. Wang, W. Zhang, and G. Atkin, "Ribosome binding model using a codebook and exponential metric," in *2007 IEEE International Conference on Electro/Information Technology*, IEEE, May 2007, pp. 438–442. DOI: 10.1109/EIT.2007.4374464.

[43] M. A. Bataineh, L. Huang, M. Alonso, N. Menhart, and G. E. Atkin, "Analysis of gene translation using a communications theory approach," in *Advances in Computational Biology*, Springer, 2010, pp. 387–397.

[44] M. Al Bataineh, *Analysis of Genomic Translation Using a Communications Theory Approach*. Illinois Institute of Technology, 2010.

[45] M. Al Bataineh, "Analysis of genomic translation using a communications theory approach," Order No. 3435813, Ph.D. Dissertation, Illinois Institute of Technology, 2010. [Online]. Available: https://uaeu.idm.oclc.org/login?url=https://www.proquest.com/dissertations-theses/analysis-genomic-translation-using-communications/docview/822457384/se-2.